Solid State Physics Practice

Budapest University of Technology and Economics

Homework — The Free Electron Gas

Fall 2026

Instructions: The deadline for this homework is **1 week**. Please upload a **scanned copy of your handwritten solution**. Show all steps of your calculations.

Notation: $E(\mathbf{k}) = \hbar^2 k^2/2m$ is the free-electron dispersion; $V = abc$ is the volume of the sample; $g(E) = \sum_{\mathbf{k},\sigma} \delta\big(E - E(\mathbf{k})\big)$ is the density of states (spin included); $n = N/V$ is the electron density. Useful numbers: $\hbar^2/2m = 3.81\,\text{eV}\,\text{Å}^2$, $k_B = 8.617 \times 10^{-5}\,\text{eV/K}$.

Exercise 1 — Free Electrons in a Box [10 points]

Consider a gas of N non-interacting electrons of mass m confined to a rectangular sample with edge lengths a, b and c along the x, y and z directions, respectively ($a \neq b \neq c$). Use periodic boundary conditions along all three directions. Unless stated otherwise, the sample is large, $a, b, c \to \infty$ in the sense that $\mathbf{k}$ can be treated as quasi-continuous when sums are converted to integrals, and the temperature is $T = 0$.

(a) Write down the single-particle Schrödinger equation and the periodic boundary conditions. Show that the normalized plane waves $\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\mathbf{r}}/\sqrt{V}$ are eigenstates, and determine the allowed wavevectors $\mathbf{k} = (k_x, k_y, k_z)$. What is the volume of k-space occupied by a single allowed $\mathbf{k}$ point? How many electron states (spin included) belong to a volume $\Omega_{\mathbf{k}}$ of k-space? **[1.5 pt]**

(b) Sketch the dispersion relation $E(k_x)$ along a line cut with fixed k_y and k_z. Mark on the sketch the allowed values of k_x and the offset of the parabola. Draw two cuts in the same figure: $(k_y, k_z) = (0, 0)$ and $(k_y, k_z) = (2\pi/b,\, 0)$, and give the energy difference between their minima. Which states on a cut are degenerate? How does the picture change as a is increased? **[1.5 pt]**

(c) Calculate the density of states $g(E)$ of the three-dimensional free electron gas, including the spin degeneracy. Start from the number of states $\mathcal{N}(E)$ with energy below E, and show that

$$g(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}.$$

What does this expression tell you about the dependence of $g(E)$ on the shape (a, b, c) of the sample? **[2 pt]**

(d) Determine the Fermi wavevector k_F and the Fermi energy E_F of N electrons at $T = 0$, and express them through the density $n = N/V$. Show that $g(E_F) = \dfrac{3N}{2E_F}$ and that the total ground-state energy is $E_0 = \frac{3}{5}NE_F$. Evaluate k_F, E_F and the Fermi temperature $T_F = E_F/k_B$ for a sample with $a = 2\,\text{mm}$, $b = 3\,\text{mm}$, $c = 5\,\text{mm}$ containing $N = 2.5 \times 10^{21}$ electrons. **[2 pt]**

(e) Consider again the ground state of part (d) and let the sample be compressed.

(i) Compute the ground-state energy as $E_0 = \int_0^{E_F} E\, g(E)\, dE$ and express it as a function of the volume V at fixed N. From $P = -\partial E_0/\partial V$ obtain the pressure of the electron gas, $P = \frac{2}{5}nE_F$, and the bulk modulus $B = -V\,\partial P/\partial V = \frac{2}{3}nE_F$. **[1 pt]**

(ii) Now compress the sample along x only, $a \to \lambda a$ with b and c unchanged ($\lambda < 1$), so slowly that no electron changes its quantum numbers (n_x, n_y, n_z). Using $\langle k_x^2\rangle = k_F^2/5$ over the Fermi sphere, show that the energy of the compressed state is $E_{\text{ad}}(\lambda) = \frac{1}{3}E_0\,(\lambda^{-2} + 2)$, where E_0 is the initial energy. Show that the force per unit area on the yz faces, $P_x = -\dfrac{1}{bc}\dfrac{\partial E_{\text{ad}}}{\partial a}$, equals the pressure of part (i) at $\lambda = 1$. **[1 pt]**

(iii) Compare E_{ad} with the energy E_{rel} of the ground state of the compressed sample of the same volume $V' = \lambda V$ (electrons redistributed over a Fermi sphere). Show that for $\lambda = 1 - \varepsilon$, $\varepsilon \ll 1$, the excess is $E_{\text{ad}} - E_{\text{rel}} = \frac{4}{9}E_0\,\varepsilon^2$, and evaluate $(E_{\text{ad}} - E_{\text{rel}})/N$ for $\varepsilon = 10^{-3}$ in the sample of part (d). What is the shape of the occupied region in $\mathbf{k}$-space in the compressed state? **[1 pt]**